

3937. Minimum Operations to Make Array Modulo Alternating I

Solved ●

Medium   Hint

You are given an integer array `nums` and an integer `k`.

In one operation, you can **increase** or **decrease** any element of `nums` by 1.

An array is called **modulo alternating** if there exist two **distinct** integers `x` and `y` ($0 \leq x, y < k$) such that:

- For every **even** index `i`, `nums[i] % k == x`
- For every **odd** index `i`, `nums[i] % k == y`

Return the **minimum** number of operations required to make `nums` **modulo alternating**.

Example 1:

Input: `nums = [1,4,2,8]`, `k = 3`

Output: 2

Explanation:

- Let's choose `x = 1` for even indices and `y = 2` for odd indices.
- Perform the following operations:
 - Increment `nums[1] = 4` by 1, giving `nums = [1, 5, 2, 8]`.
 - Decrement `nums[2] = 2` by 1, giving `nums = [1, 5, 1, 8]`.
- Now, for even indices, `nums[i] % k = 1`, and for odd indices, `nums[i] % k = 2`.
- Thus, the total number of operations required is 2.

Example 2:

Input: `nums = [1,1,1]`, `k = 3`

Output: 1

Explanation:

- Incrementing `nums[1]` by 1 gives `nums = [1, 2, 1]`, which satisfies the condition with `x = 1` and `y = 2`.
- Thus, the total number of operations required is 1.

Constraints:

- $1 \leq \text{nums.length} \leq 100$
- $1 \leq \text{nums}[i] \leq 10^9$
- $2 \leq k \leq 100$

Seen this question in a real interview before? 1/6

Yes No

Accepted 16,791/36K | Acceptance Rate 46.6%

Topics	▼
	▼
Hint 1	▼
Hint 2	▼
Discussion (23)	▼